

CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL, COIMBATORE

SAMPLE QUESTION PAPER CLASS XI (IB) PHYSICS

SECTION A

01. Keshav takes a ride on the giant wheel of radius 'R' making 5 rounds. What is his displacement at the end of the ride?

- a. $2\pi R$
- b. $10\pi R$
- c. Zero

ANSWER	
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02. The formation of rainbow is a proof of the following:

- a. the earth is spherical in shape
- b. sunlight has different colors
- c. rain drops are spherical in shape

ANSWER	
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03. Which of the following principles is involved in the generation of electricity?

- a. Electromagnetic induction
- b. Coulomb's law
- c. Law of conservation of charge

ANSWER	
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04. Which of the following elements are isotopes?

Element	Mass number	Atomic number
A	56	32
B	56	31
C	57	31
D	57	33

- a. A, B
- b. C, D
- c. B, C

ANSWER	
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05. Light-year is a unit of

- a. distance
- b. time
- c. light intensity

ANSWER	
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SAMPLE QUESTION PAPER
SECTION B

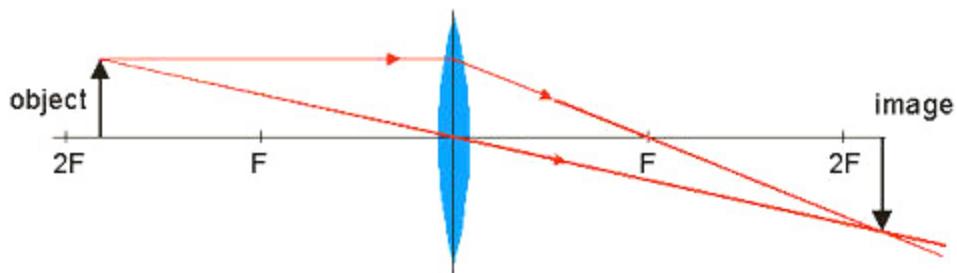
01. A boy of 40 Kg tries to push a large stone of 1200 Kg, but the stone does not move. Find the work done in pushing the stone. Justify your answer.
02. The acceleration of an object moving with a uniform speed in a circular path is zero. Is it **True or False? Justify.**
03. A convex lens forms an image of a distant star at a distance of 50cm from the lens. What is the focal length of the lens?
04. A Uranium nucleus (mass number 235) undergoes beta decay. What will be its mass number after the decay?
05. Which of the following is not a part of the electromagnetic waves.
 - a. Ultrasonic waves
 - b. Gamma rays
 - c. Radio waves

Ans	
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06. What is the quantity for which Horsepower is the unit? Name the SI unit for this quantity.

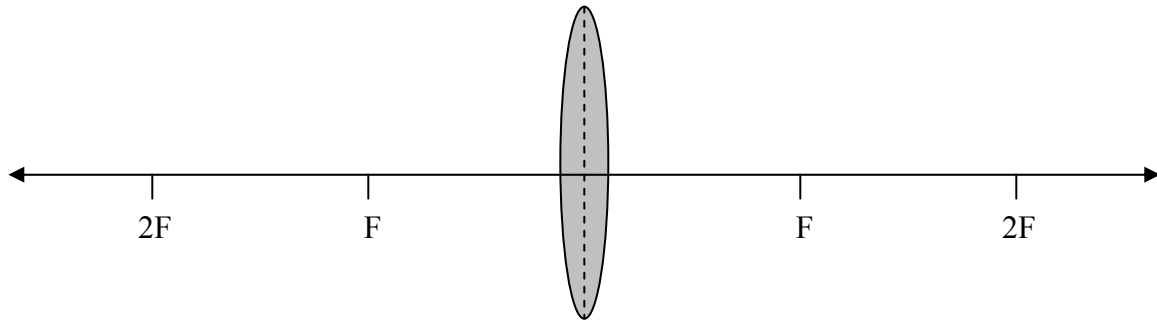
SECTION C

01. Mukesh is standing at the balcony of his 5th floor apartment that is at a height of 45m from the ground. Rahul is standing directly below the balcony in the ground. Suddenly Mukesh drops his watch by mistake. How much time later would Rahul catch the watch? What is the speed with which the watch falls into Rahul's hand? (take $g=10\text{m/s}^2$).
02. The following figure shows the image formed by a lens when the object is between F and 2F.

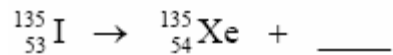


- a. What is the nature of this image?
- b. Draw the ray diagram showing the positions of the object and the image in order to form an enlarged, erect, virtual image of the object.

SAMPLE QUESTION PAPER



03. The following is a nuclear reaction:



(a) What does the number 135 indicate?

(b) Complete the reaction.

04. The following technical specification is written on the back panel of an electrical appliance. Rewrite the same with the appropriate units.

220-240...., AC 50/60, 2.5....